EMS611U: Conceptual Design of Aircraft

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Week 8

Thrust-to-Weight Ratio & Wing Loading

Introduction

Thrust-to-weight Ratio
T/W

Wing Loading
W/S

- These are the two parameters that affect the aircraft performance the most.
- Optimization of these parameters forms a major part of the analytical design activities conducted after an initial design layout.

We need a credible initial estimate.

Wing Loading
W/S

Low → Large wing → Enough volume for landing gear & fuel

High → Small wing → Not enough volume for landing gear & fuel
placement in the fuselage
increment of wetted area and additional drag force

Introduction

$\frac{T}{W}, \frac{W}{S}$ → Take-off distance

For short take-off

- Large wing, small engine; $\frac{W}{S}\downarrow$ $\frac{T}{W}\downarrow$ Smaller "Thrust"
- Small wing, large engine; $\frac{W}{S}\uparrow$ $\frac{T}{W}\uparrow$ Higher "Thrust"

- It is frequently difficult to use historical data to independently select initial values for T/W and W/S

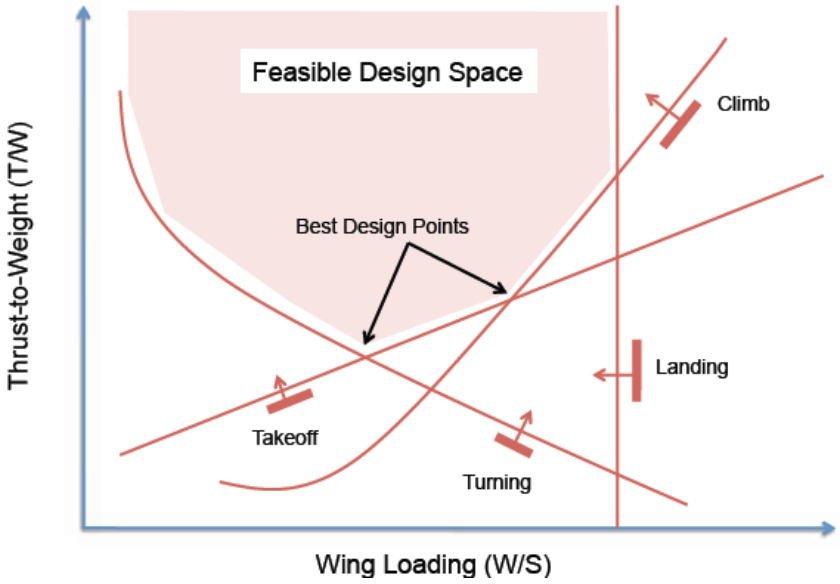
Introduction

- The designer must guess at one of the parameters and use that guess to calculate the parameter from the critical design parameters.

$\frac{W}{S}$ → Critical design requirement: **"stall speed"** during the approach for landing.

(Approach stall speed is independent of engine size)

Then, the estimated W/S can be used to calculate the T/W required to attain other performance drivers such as **"single-engine rate of climb"**



Thrust-to-Weight Ratio

Thrust-to-Weight Definitions

$\frac{T}{W} \uparrow$

- Quick acceleration
- Rapid climb
- Higher maximum speed
- Higher turn rates

- More fuel consumption
- Higher engine weight

❖ $W \downarrow$ (fuel burning) $\rightarrow \frac{T}{W} \neq const$

❖ $T = T(h, V)$

- Design $\boldsymbol{T/W}$ **during sea-level (zero-velocity)**, **standard-day conditions** at **design take-off weight** and **maximum throttle setting**.

- If a required T/W is calculated at some other condition, it must be adjusted back to take-off conditions.

Thrust-to-Weight Ratio

Power Loading and Horsepower to Weight

Jet engine aircraft $\rightarrow \frac{T}{W}$

Propeller-powered aircraft → "power loading" $\frac{W}{hp}$

Power loading for "most aircraft": $10 - 15\ [lb/hp]$

Power loading for "aerobatic aircraft": $6\ [lb/hp]$

Equivalent T/W for propellered aircraft:

$$\frac{T}{W} = \left(\frac{\eta_p}{V}\right)\left(\frac{P}{W}\right) = \left(\frac{550\,\eta_p}{V}\right)\left(\frac{hp}{W}\right)\ [fps\ units]$$

Thrust-to-Weight Ratio

Statistical Estimation of T/W

Aircraft type	Typical installed T/W
Jet trainer	0.4
Jet fighter (dogfighter)	0.9
Jet fighter (other)	0.6
Military cargo/bomber	0.25
Jet transport	0.25

At maximum power settings at sea-level and zero-velocity (static) condition.

Aircraft type	Typical hp/W	Typical power loading (W/hp)
Powered sailplane	0.04	25
Homebuilt	0.08	12
General aviation—single engine	0.07	14
General aviation—twin engine	0.17	6
Agricultural	0.09	11
Twin turboprop	0.20	5
Flying boat	0.10	10

Thrust-to-Weight Ratio

At take-off weights: Modern jet fighter → $T/W \cong 1.0$

At combat conditions (With fuel burning $W \downarrow$): $T/W > 1.0$ capable of accelerating

$T/W_0 = a\ M_{max}^C$	a	C
Jet trainer	0.488	0.728
Jet fighter (dogfighter)	0.648	0.594
Jet fighter (other)	0.514	0.141
Military cargo/bomber	0.244	0.341
Jet transport	0.267	0.363

$hp/W_0 = a\ V_{max}^C$	a	C
Sailplane—powered	0.043	0
Homebuilt—metal/wood	0.005	0.57
Homebuilt—composite	0.004	0.57
General aviation—single engine	0.024	0.22
General aviation—twin engine	0.034	0.32
Agricultural aircraft	0.008	0.50
Twin turboprop	0.012	0.50
Flying boat	0.029	0.23

Curve-fit equations based upon maximum Mach number or velocity. These can be used for first estimation. Units: Vmax in mph, hp/W0 in hp/lb (Mmax is dimensionless).

Thrust-to-Weight Ratio

Thrust Matching

For aircraft designed primarily for "efficiency during cruise":

- A better initial estimate of the required T/W can be obtained by "thrust matching":
- Comparison of selected engine's thrust available during cruise to the estimated aircraft drag.

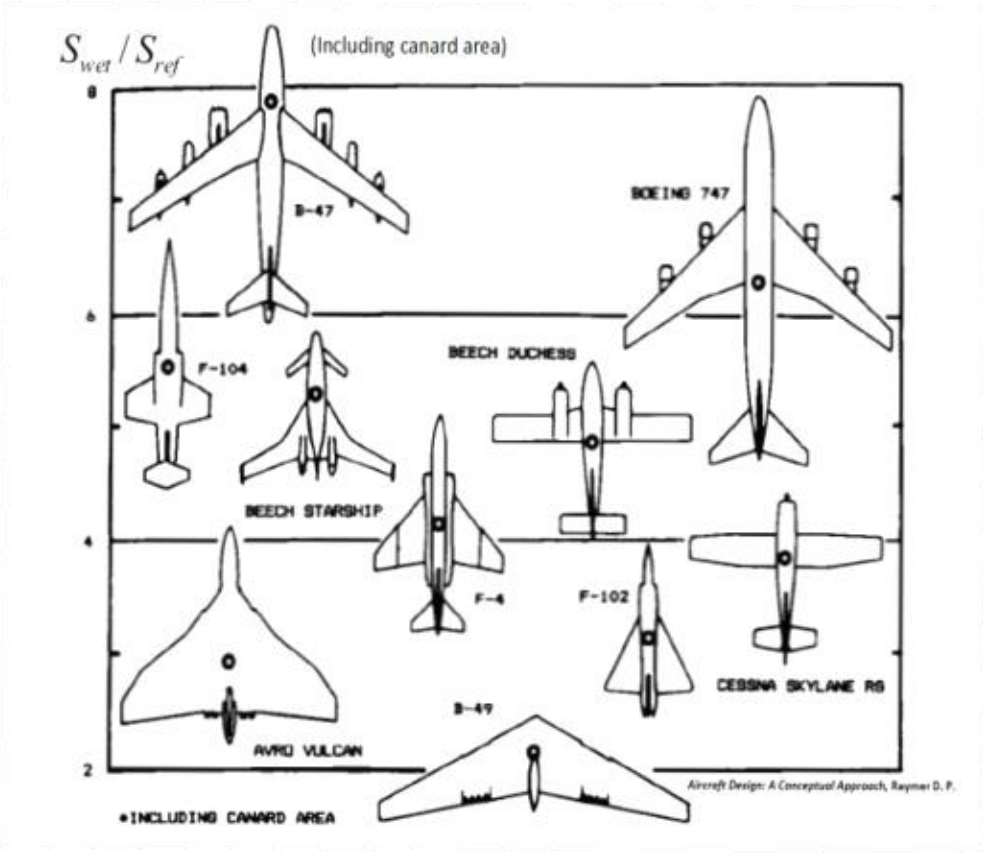
Cruise flight condition:

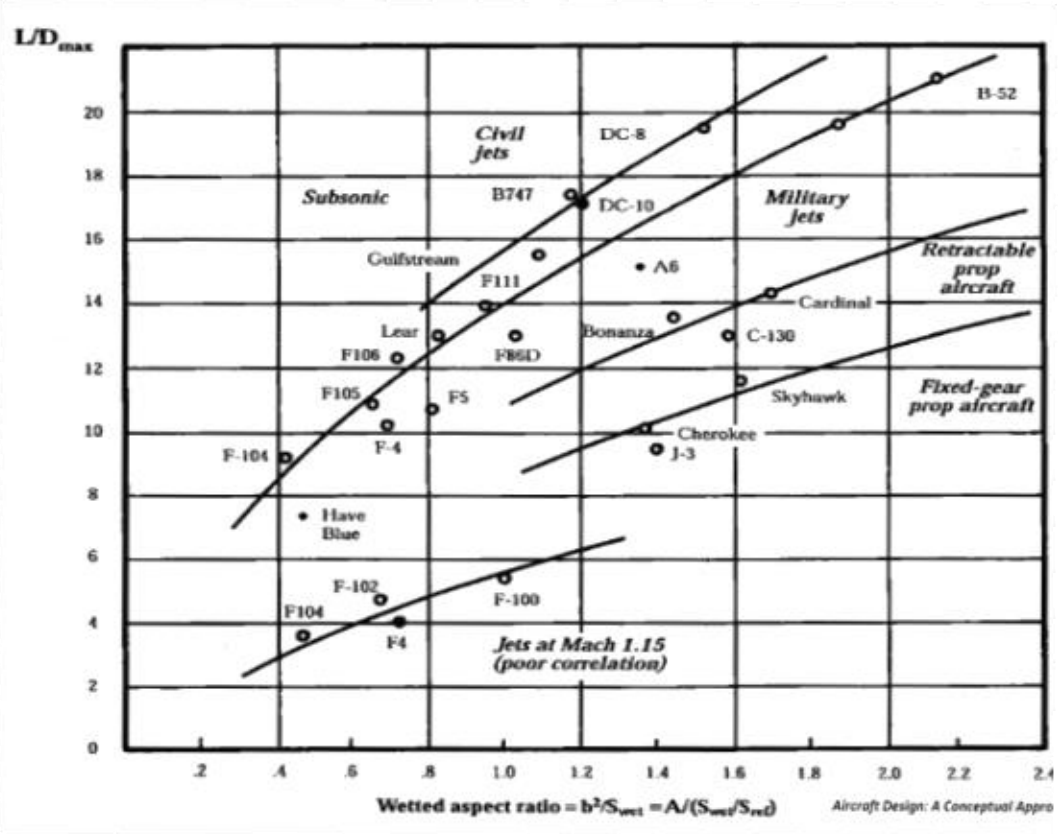
$$\left.\begin{aligned} L &= W \\ T &= D \end{aligned}\right\} \quad \left(\frac{T}{W}\right)_{cruise} = \left(\frac{D}{L}\right)_{cruise} = \frac{1}{(L/D)_{cruise}}$$

$\longrightarrow$ *Estimated value can be used.*

This T/W is at cruise condition. T/W, for take-off at sea-level condition, must be calculated.

Thrust-to-Weight Ratio





	Cruise
Jet	$0.866\,(L/D)_{max}$
Prop	$(L/D)_{max}$

Thrust-to-Weight Ratio

Thrust Matching: Assumptions and Weight at the Start of Cruise

- Assumes the aircraft cruises at about the optimum altitude for the as-yet-unknown W/S.
- Invalid if the mission forces another altitude, e.g. sea level.
- After W/S is selected, recompute L/D at the actual cruise conditions and recheck T/W.
- The highest cruise weight is at the beginning of the cruise: fuel has already been burned in take-off and climb.

Mission segment	W_i/W_{i-1}
Warm-up and take-off	0.970
Climb	0.985
Landing	0.995

Typical mission weight fractions (Table 3.2)

$$W_{\text{cruise start}} = 0.970 \times 0.985\, W_{TO} \approx 0.956\, W_{TO}$$

Thrust-to-Weight Ratio

Thrust Matching: Thrust Lapse with Altitude (Jet Engines)

- Cruise thrust is lower than sea-level static take-off thrust: $T_{\text{cruise}} < T_{\text{TO}}$
- If thrust falls with altitude, why not cruise low, where the engine is strong? There are two reasons:
  - The thrust needed doesn't change. In level flight, thrust = drag, and drag = W/(L/D). That depends on the aircraft's weight and L/D, not on altitude.
  - The engine burns less fuel per unit of thrust up high. This is the SFC (specific fuel consumption). Cold air at altitude improves engine efficiency, so SFC is lower. The thin air also means the aircraft must fly faster to get the same lift. It covers more distance per hour, and the thrust needed stays the same. Both effects favour cruising high.
- Jets cruise where SFC is lowest, typically 30,000–40,000 ft (≈ 10,000 m). SFC improves, but thrust falls with air density. The engine is sized at the throttle setting with the best SFC: usually 70–100% of maximum continuous (non-afterburning) thrust.
- So the cruise requirement must be converted to the equivalent **take-off T/W**.

Thrust-to-Weight Ratio

Thrust Lapse at Cruise by Engine Type

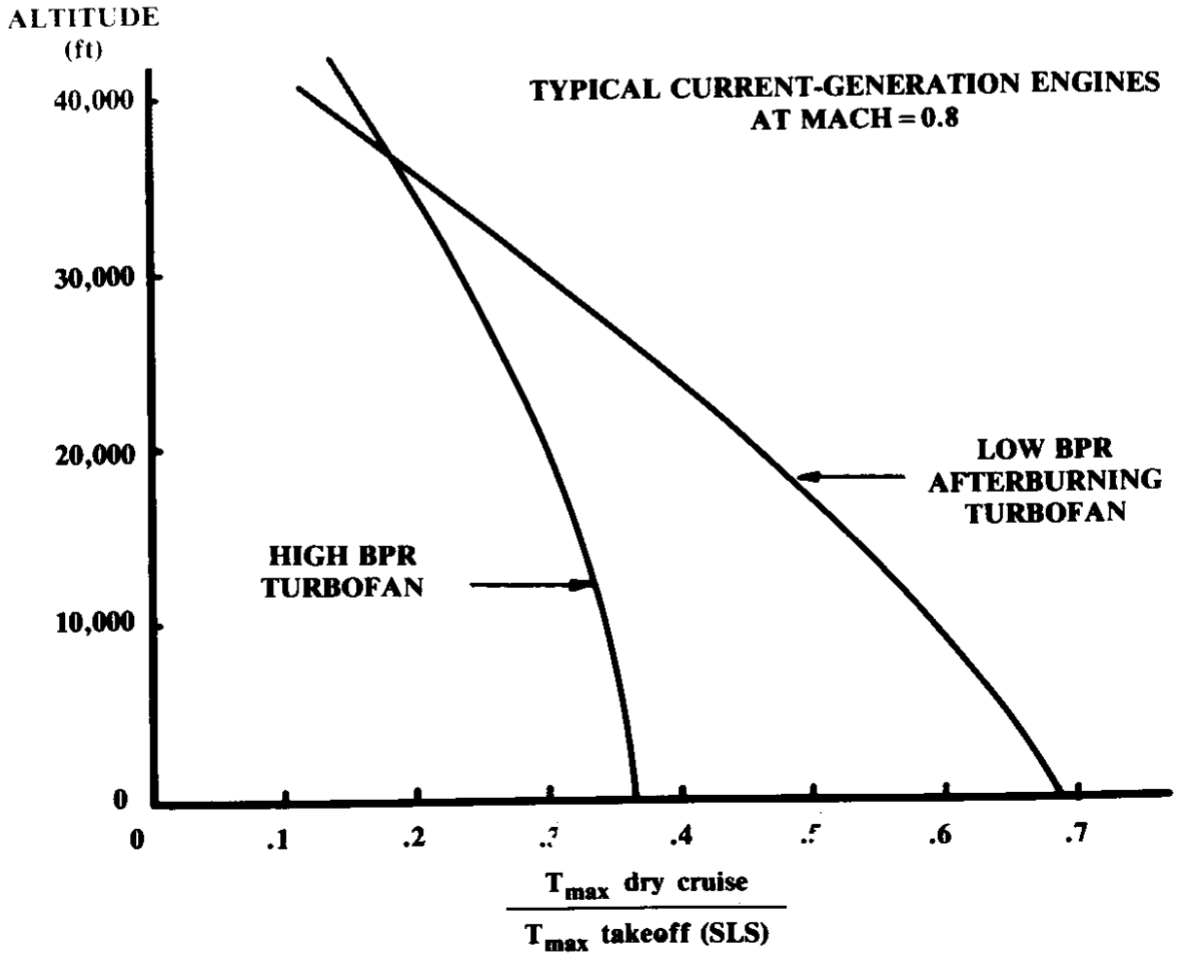


Fig. 5.1 Thrust lapse at cruise (Raymer)

High-bypass turbofan (transport): cruise thrust ≈ **20–25%** of take-off thrust

Low-bypass afterburning turbofan / turbojet: cruise thrust ≈ **40–70%** of take-off thrust

Thrust Matching: Piston and Turboprop Engines

- **Piston engine:** power depends on the density of the air reaching the manifold: without a supercharger it falls with $\sigma = \rho/\rho_{SL}$.
- A non-supercharged engine at 10,000 ft (3,048 m) has only about **73%** of its sea-level power.
- A supercharger or turbocharger holds the manifold density up to its limit (Fig. 5.2).
- Piston-propeller aircraft cruise at about **75%** of take-off power.

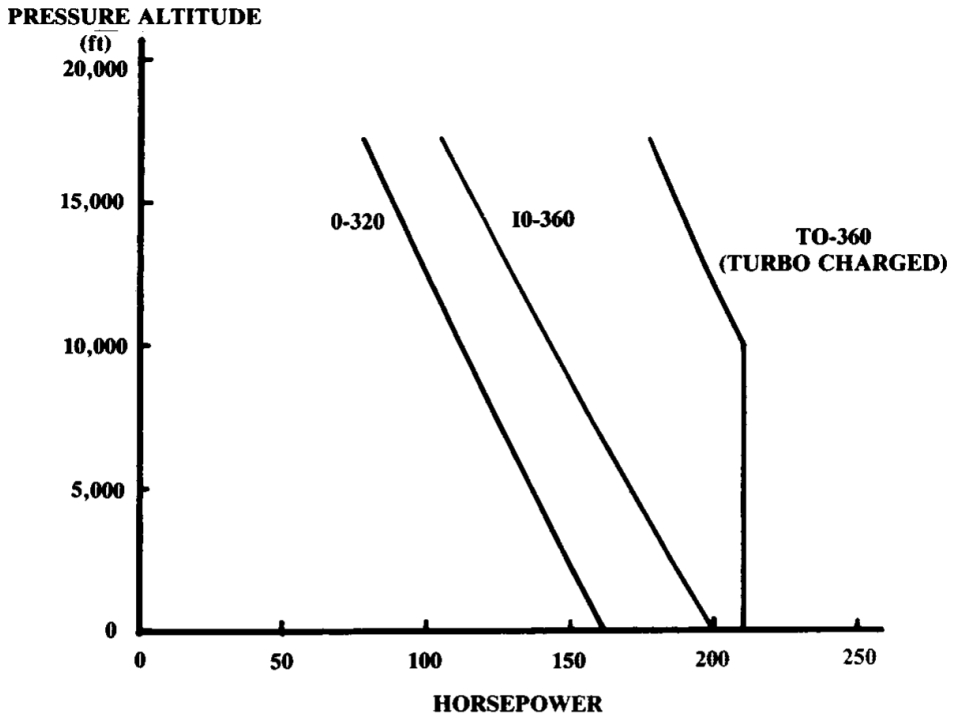


Fig. 5.2 Piston engine power vs altitude (Raymer)

Turboprop: residual exhaust thrust is converted to horsepower: **equivalent shaft horsepower (eshp)**. Cruise eshp ≈ **60–80%** of take-off.

Thrust-to-Weight Ratio

Adjusting the Cruise Requirement Back to Take-off Conditions

Take-off T/W needed to satisfy the cruise (thrust-matching) condition [Eq. 5.3]:

$$\left(\frac{T}{W}\right)_{TO} = \left(\frac{T}{W}\right)_{\text{cruise}} \left(\frac{W_{\text{cruise}}}{W_{TO}}\right) \left(\frac{T_{TO}}{T_{\text{cruise}}}\right)$$

1st factor: cruise requirement, 1/(L/D) in cruise

2nd factor: weight ratio, ≈ 0.956 (or from mission data)

3rd factor: 1 / thrust lapse, from engine data (e.g. Fig. 5.1)

Thrust-to-Weight Ratio

Propeller Aircraft (Eq. 5.4) and a Quick Check

From:

$$\frac{T}{W} = \left(\frac{\eta_p}{V}\right)\left(\frac{P}{W}\right) = \left(\frac{550\,\eta_p}{V}\right)\left(\frac{hp}{W}\right) \quad [fps\ units]$$

$$\left(\frac{hp}{W}\right)_{TO} = \left(\frac{V_{\text{cruise}}}{550\,\eta_p}\right)\left(\frac{1}{(L/D)_{\text{cruise}}}\right)\left(\frac{W_{\text{cruise}}}{W_{TO}}\right)\left(\frac{hp_{TO}}{hp_{\text{cruise}}}\right)$$

Typically $\eta_p = 0.8$:

Illustration (jet transport, my numbers): $(L/D)_{\text{cruise}} = 17$, $W_{\text{cruise}}/W_{\text{TO}} = 0.956$, cruise thrust = 22.5% of take-off thrust

$\left(\frac{T}{W}\right)_{TO} = \frac{1}{17} \times 0.956 \times \frac{1}{0.225} \approx 0.25$ ← consistent with Table 5.1 (jet transport: 0.25)

Thrust-to-Weight Ratio

T/W is Often Set by a Climb Requirement

- T/W is often determined by a climb requirement rather than by cruise.
- If climb needs a lot of thrust, the engines must be throttled way back in cruise, where an aircraft engine (especially a jet) is very inefficient.

$$\left(\frac{T}{W}\right)_{\text{cruise}} = \frac{1}{(L/D)_{\text{cruise}}}$$

level flight: thrust only balances drag

In a climb the engine has two jobs: **balance the drag** and **lift the weight up the slope**.

$$\left(\frac{T}{W}\right)_{\text{climb}} = \frac{1}{(L/D)_{\text{climb}}} + \frac{V_{\text{vertical}}}{V}$$

+ climb gradient

Thrust-to-Weight Ratio

The Extra Thrust is the Climb Gradient

The extra thrust, as a fraction of the weight, is the **climb gradient** = sin γ ≈ V_{vertical} / V.

Illustration (my numbers): a twin-engined transport must climb on one engine after an engine failure, with a gradient of at least 2.4% (FAR 25). With L/D ≈ 10: the surviving engine supplies 0.1 + 0.024 = 0.124 W, so both engines give about **0.25**, though cruise needs only about 0.06.

- First pass: select the **higher** of the statistical value (Tables 5.1–5.4) and the thrust-matching value.
- After W/S is selected, recheck T/W against all requirements.

Wing Loading

Road Map: Every Requirement Gives a Limit on W/S

Stall speed	Take-off distance	Catapult take-off
Landing distance	Arrested landing	Cruise (best range)
Loiter (endurance)	Turning (instantaneous, sustained)	Climb, glide and ceiling

Select the **lowest** W/S (all converted to take-off conditions), then **recheck T/W**

Wing Loading

Wing Loading and Why it Matters

$$\frac{W}{S} = \frac{W_{TO}}{S_{\text{ref}}}$$

- Weight of the aircraft divided by the area of the reference wing: the whole wing, including the part inside the fuselage.

- "Wing loading" normally means take-off wing loading; it can also refer to combat or other flight conditions.
- Physically, weight ÷ area is a **pressure**: the average pressure difference between the lower and upper surfaces must equal W/S.

Illustration: jet transport 120 lb/ft² ≈ 5.7 kPa ≈ **6%** of an atmosphere; sailplane ≈ 0.3 kPa

Wing Loading: What it Affects, and the Cost of a Bigger Wing

W/S affects:

- Stall speed
- Climb rate
- Take-off distance
- Landing distance
- Turn performance
- Design lift coefficient
- Wetted area and wing span → drag

Larger wing (lower W/S): performance improves, but the **extra drag and empty weight** raise take-off weight. The sizing equation magnifies this: the weight increase is more than proportional.

$\frac{W}{S} \downarrow \rightarrow$ bigger wing $\rightarrow$ **performance** $\uparrow$, **drag** $\uparrow$ **and** $W_{TO} \uparrow$

Wing Loading

Typical Wing Loadings (Table 5.5)

Aircraft type	W/S [lb/ft²]	W/S [kg/m²]
Sailplane	6	30
Homebuilt	11	54
General aviation – single engine	17	83
General aviation – twin engine	26	127
Twin turboprop	40	195
Jet trainer	50	244
Jet fighter	70	342
Jet transport / bomber	120	586

- The faster the aircraft, the higher the wing loading: each m² of wing can carry more lift at higher dynamic pressure.
- Sailplane to jet transport: a factor of **20**.

×4.88 for kg/m², ×47.9 for N/m²

Wing Loading

Selecting Wing Loading: Use the Lowest Value, then Question It

- T/W and W/S must be optimised together; estimates only need to be good enough that the layout will not need a complete revision.
- So that the wing gives enough lift in all circumstances, select the **lowest** of the estimated wing loadings.
- If an unreasonably low W/S is driven by only one requirement, find another way to meet it:

Stall speed requirement too demanding	→ Equip a **high-lift flap system** (higher $C_{L_{\max}}$)
Take-off distance or climb rate too demanding	→ **Increase T/W** (bigger engine) instead of a bigger wing

Wing Loading: Stall Speed

Why Stall Speed Matters

- Stall speed is set directly by W/S and $C_{L_{\max}}$.
- A major contributor to flight safety: many fatal accidents each year are due to "failure to maintain flying speed".
- The approach speed (the most important factor in landing distance and in post-touchdown accidents) is defined by the stall speed.

Wing Loading: Stall Speed

Stall Speed Requirements and Approach Speed

- **FAR 23** aircraft (below 12,500 lb ≈ 5,670 kg): $V_{\text{stall}} \leq 61\ \text{kt}$ (113 km/h).
- Trainers / low-time pilots: about **50 kt** (93 km/h) is the upper limit.

Application	V_{approach}
Civil	$\geq 1.3\ V_{\text{stall}}$
Military	$\geq 1.2\ V_{\text{stall}}$
Carrier-based	$1.15\ V_{\text{stall}}$

Required stall speed = approach speed ÷ **1.3**, **1.2** or **1.15**

Wing Loading: Stall Speed

Stall Speed from Wing Loading and Maximum Lift Coefficient

Level flight: lift = weight. At stall the wing works at its maximum lift coefficient:

$$W = L = q_{\text{stall}}\, S\, C_{L_{\max}} = \frac{1}{2}\rho V_{\text{stall}}^2\, S\, C_{L_{\max}} \qquad (5.5)$$

Dividing by S:

$$\frac{W}{S} = \frac{1}{2}\rho V_{\text{stall}}^2\, C_{L_{\max}} \qquad (5.6)$$

Illustration: FAR 23 aircraft, $V_{\text{stall}} = 61\ \text{kt} = 103\ \text{ft/s}$, $C_{L_{\max}} = 1.6$

$$\frac{W}{S} = \frac{1}{2}(0.00238)(103)^2(1.6) \approx 20\ \text{lb/ft}^2$$

Wing Loading: Stall Speed

Maximum Lift Coefficient: Typical Values and What Drives Them

Configuration	C_{Lmax}
Plain wing, no flaps	1.2 – 1.5
Flaps on the inner part of the wing	1.6 – 2.0
Regular transport: flaps and slats	2.4
STOL (short take-off and landing)	3.0
Large flaps in propwash / jetwash	up to 5.0

Take-off $C_{L_{\max}}$ ≈ 80% of the landing value: full flap drag would hurt acceleration and climb.

$C_{L_{\max}}$ depends on:

- ✓ wing geometry and airfoil shape
- ✓ flap geometry and span
- ✓ leading-edge slot / slat
- ✓ Reynolds number, surface
- ✓ interference: fuselage, nacelles
- ✓ tail trim force (up or down)
- ✓ propwash / jetwash on the wing

Wing Loading: Stall Speed

Maximum Lift of the Wing

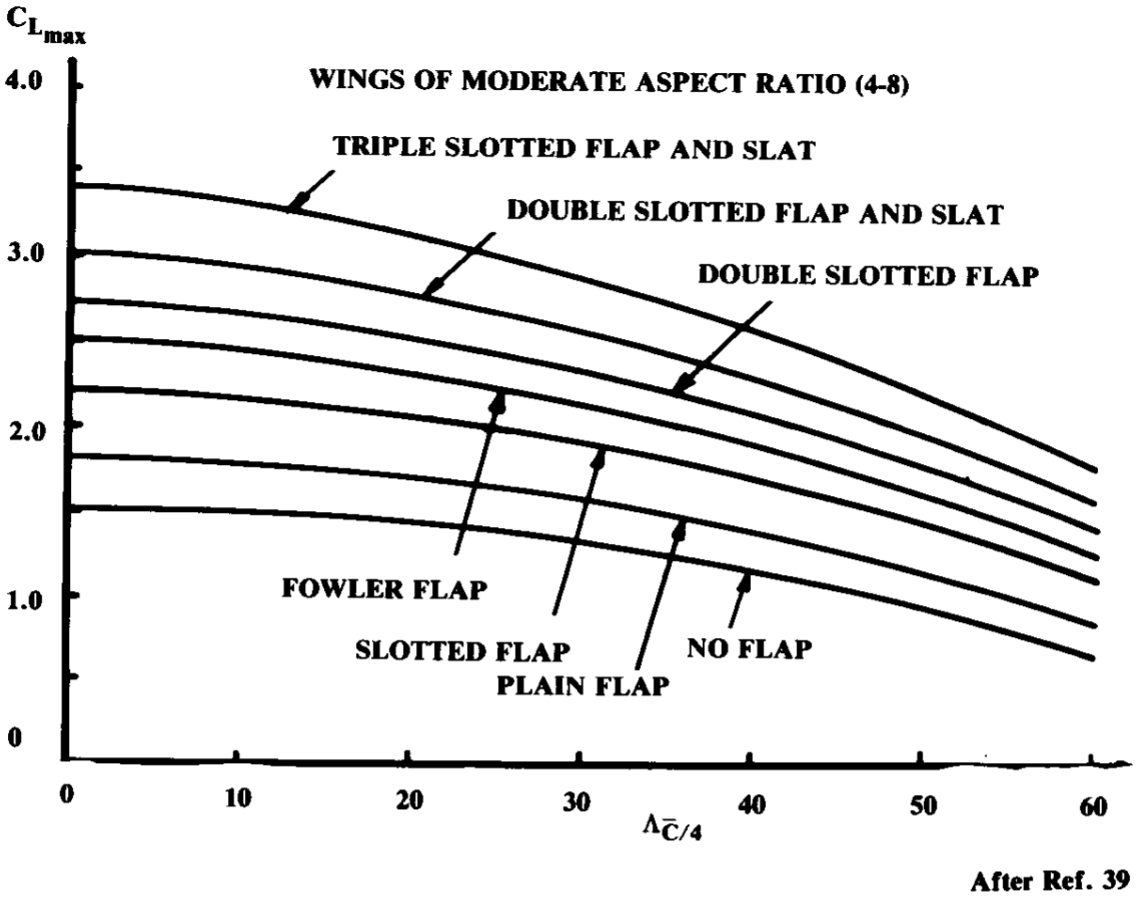


Fig. 5.3 Maximum lift coefficient (Raymer)

- Wings of moderate aspect ratio (4–8): $C_{L_{\max}}$ falls with sweep and rises with more elaborate flaps and slats.
- For A > 5 the wing $C_{L_{\max}}$ is about **90%** of the airfoil $C_{l\max}$ at the same Reynolds number.

Wing Loading: Stall Speed

Maximum Lift of the Wing: Partial-span Flaps

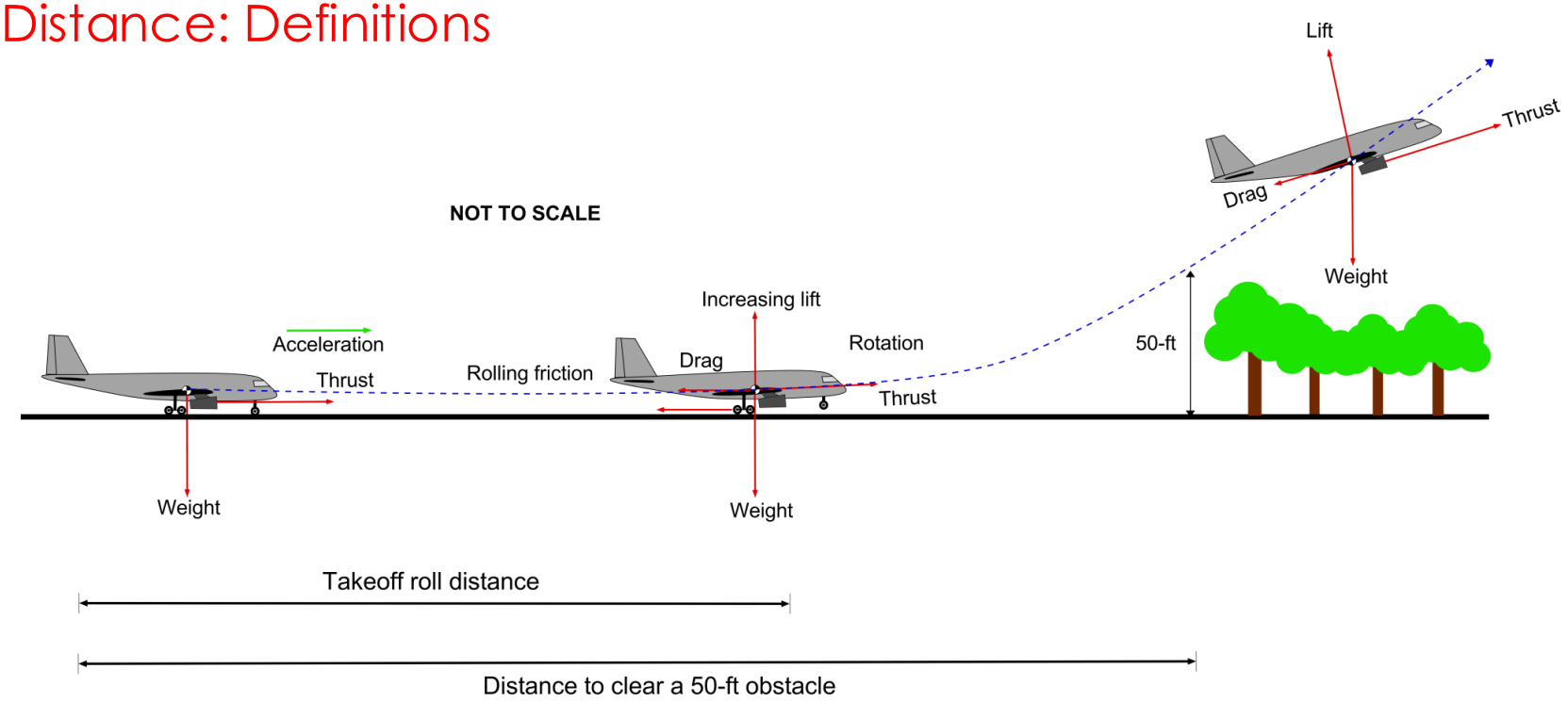
- Partial-span flaps add a discontinuous twist, changing the lift distribution and the induced downwash: the effective angle of attack varies along the span.

$$C_{L_{\max}} \cong 0.9\left\{(C_{l_{\max}})_{\text{flapped}}\frac{S_{\text{flapped}}}{S_{\text{ref}}} + (C_l)_{\text{unflapped}}\frac{S_{\text{unflapped}}}{S_{\text{ref}}}\right\}$$

(C_l)unflapped: lift coefficient of the unflapped airfoil at the angle of attack where the flapped part stalls.

Wing Loading: Take-off Distance

Take-off Distance: Definitions

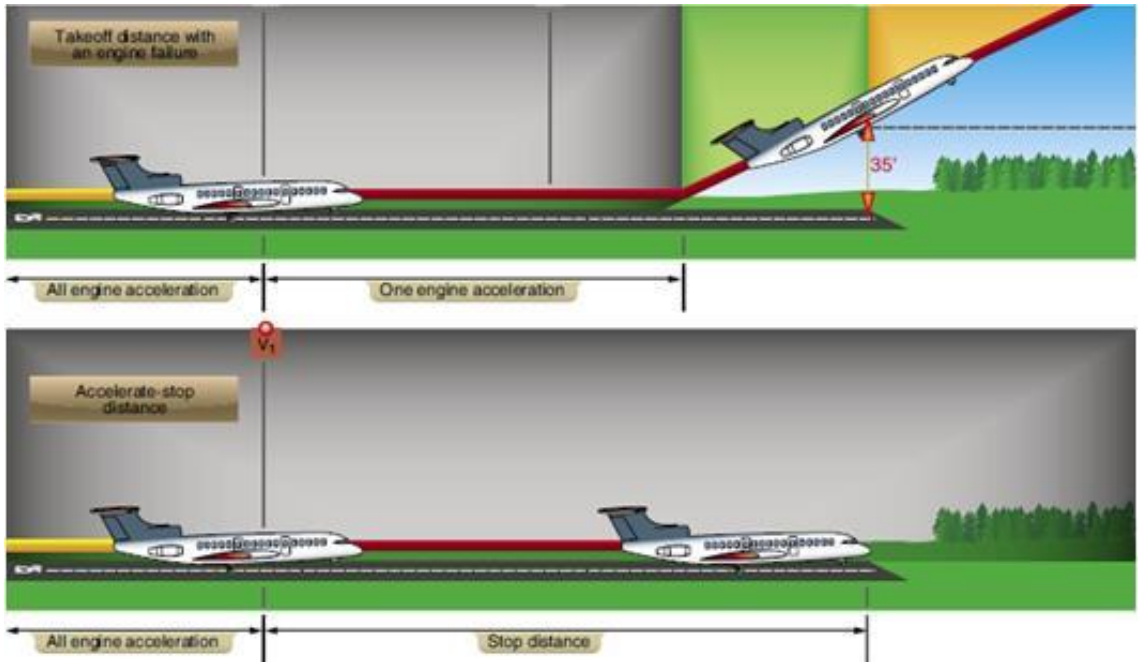


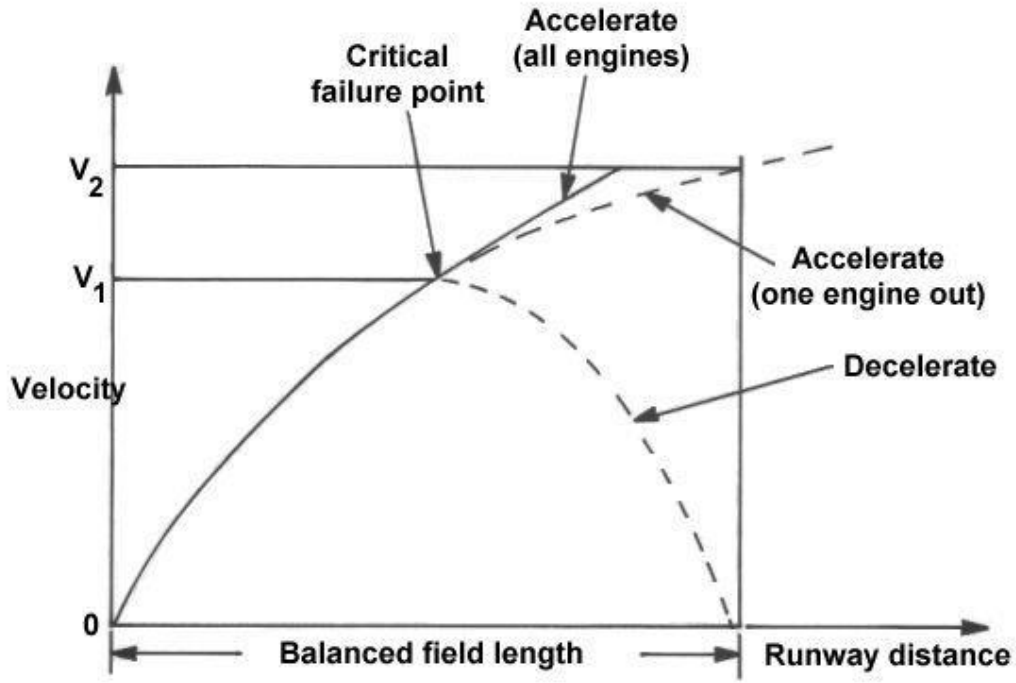
- **Ground roll:** the actual distance travelled before the wheels leave the ground.
- **Lift-off speed**, normal take-off: $V_{LOF} = 1.1\,V_{\text{stall}}$
- **Obstacle clearance distance:** from brake release until a specified height is reached:
  - **50 ft (15.24 m)**: military or small civil
  - **35 ft (10.7 m)**: commercial

Wing Loading: Take-off Distance

Balanced Field Length and Decision Speed

- **Balanced field length:** field length needed for safety if an engine fails at the worst moment (multi-engine aircraft).
- **Decision speed V_1:** the speed at which the distance to stop after an engine failure equals the distance to continue on the remaining engines.





Before V_1: stop. After V_1: continue the take-off.

Wing Loading: Take-off Distance

What Controls the Ground Roll? Factors

- Both W/S and T/W contribute to the take-off distance.
- Also: aerodynamic drag on the ground (nose lifted too early!) and rolling resistance μ, which depends on the runway surface and on tyre type, pressure and arrangement.
- Thin, high-pressure tyres on soft dirt: so much rolling resistance that the aircraft may not move. Large low-pressure tyres roll better but add drag if not retracted.

Wing Loading: Take-off Distance

Ground Roll: A Simple Physical Model

Simplest model: **constant acceleration** from thrust alone (drag and rolling resistance ignored).

$$V_{LOF}^2 = (1.1)^2 V_{\text{stall}}^2 = 1.21 \cdot \frac{2\,(W/S)}{\rho\, C_{L_{\max}}}$$

$$a \approx g\,\frac{T}{W}$$

$$S_G = \frac{V_{LOF}^2}{2a} = \frac{W/S}{\rho\, g\, C_{L_{TO}}\,(T/W)}$$

$$C_{L_{TO}} = \frac{C_{L_{\max}}}{1.21}$$

S_G ∝ **W/S ÷ (σ $C_{L_{TO}}$ T/W)** = the **take-off parameter (TOP)**

Wing Loading: Take-off Distance

Take-off Parameter (TOP): Jets and Propeller Aircraft

- Statistical correlation of take-off distance with the **take-off parameter**:

$$TOP = \frac{W/S}{\sigma \, C_{L_{TO}} \, (T/W)}$$

jet aircraft

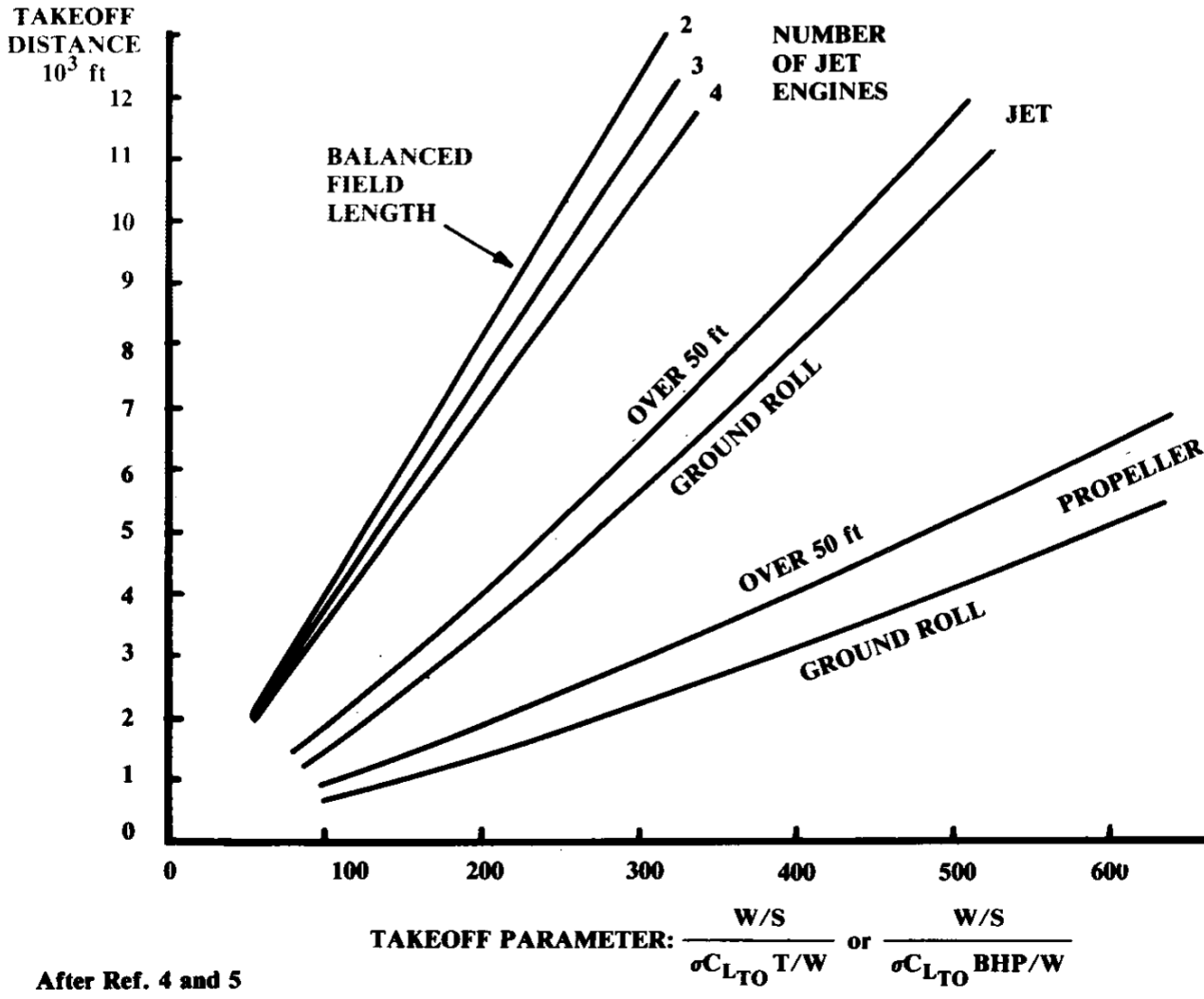
$$TOP = \frac{W/S}{\sigma \, C_{L_{TO}} \, (\mathrm{bhp}/W)}$$

propeller aircraft

Wing Loading: Take-off Distance

Take-off Distance from TOP

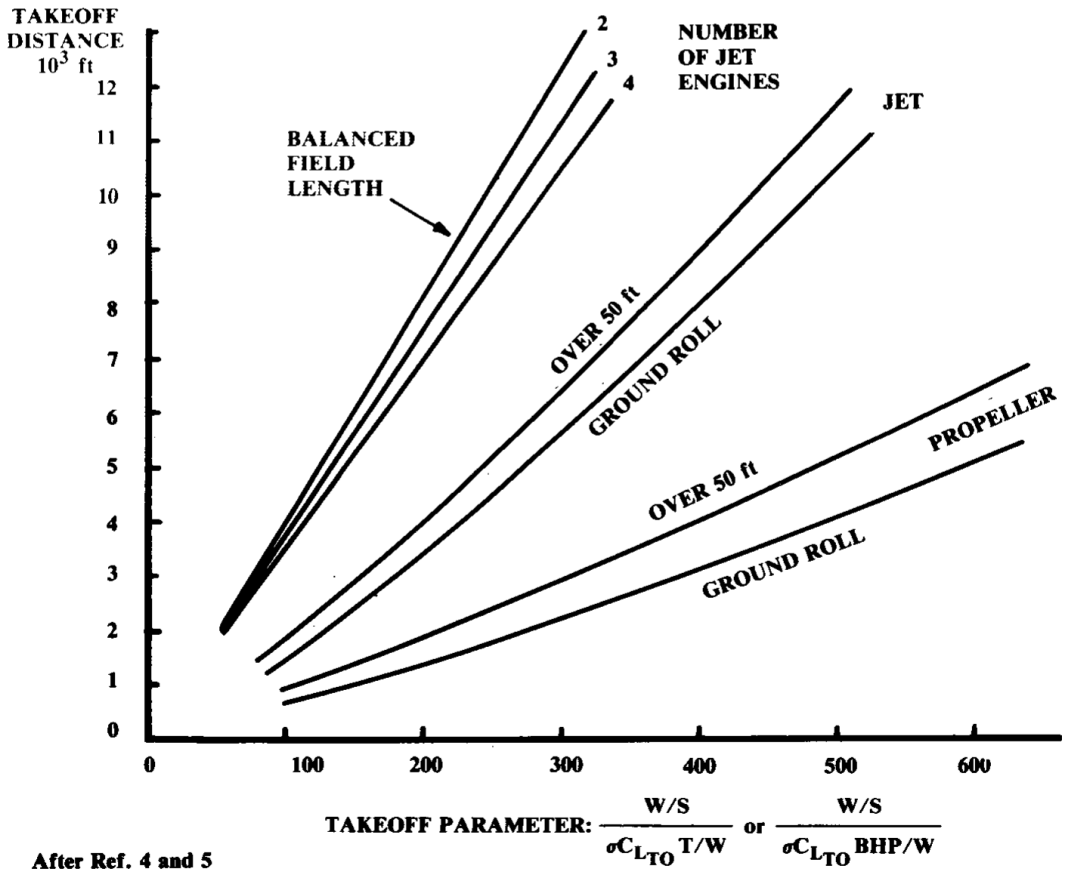
- $\sigma = \rho/\rho_{SL}$: density ratio, 1 at sea level.
- $C_{L_{TO}}$ is the actual take-off lift coefficient, not $C_{L_{\max}}$: lift-off at 1.1 $V_{\text{stall}} \rightarrow C_{L_{TO}} = C_{L_{\max}}/1.21$
- Chart (linear axes): ground roll and distance over a 50-ft obstacle for jets and propellers; balanced field length for 2-, 3- and 4-engine jets.



Wing Loading: Take-off Distance

Required Wing Loading for a Given Take-off Distance

- Read TOP from Fig. 5.4 for the required distance. The maximum allowable wing loading is then:



After Ref. 4 and 5

propeller aircraft

$$\frac{W}{S} = (TOP)\,\sigma\, C_{L_{TO}} \left(\frac{hp}{W}\right) \quad (5.8)$$

jet aircraft

$$\frac{W}{S} = (TOP)\,\sigma\, C_{L_{TO}} \left(\frac{T}{W}\right) \quad (5.9)$$

Wing Loading: Take-off Distance

Take-off Distance: How to Use the Result

- A larger T/W lets you accept a higher W/S (smaller wing) for the same runway.
- Hot day or high airfield → smaller σ → lower allowable W/S (or a longer runway).
- The equations can be solved for T/W instead if W/S is already fixed.

Note: $C_{L_{TO}}$ may also be limited by the maximum tail-down angle the landing gear allows (typically not more than 15°).

Wing Loading: Catapult Take-off

Catapult Take-off (Carrier-based Aircraft)

- Naval aircraft operate from carriers: a steam catapult accelerates the aircraft to flying speed in a very short distance.
- The catapult force is limited by steam pressure: **a light aircraft leaves at a higher speed than a heavy one**.

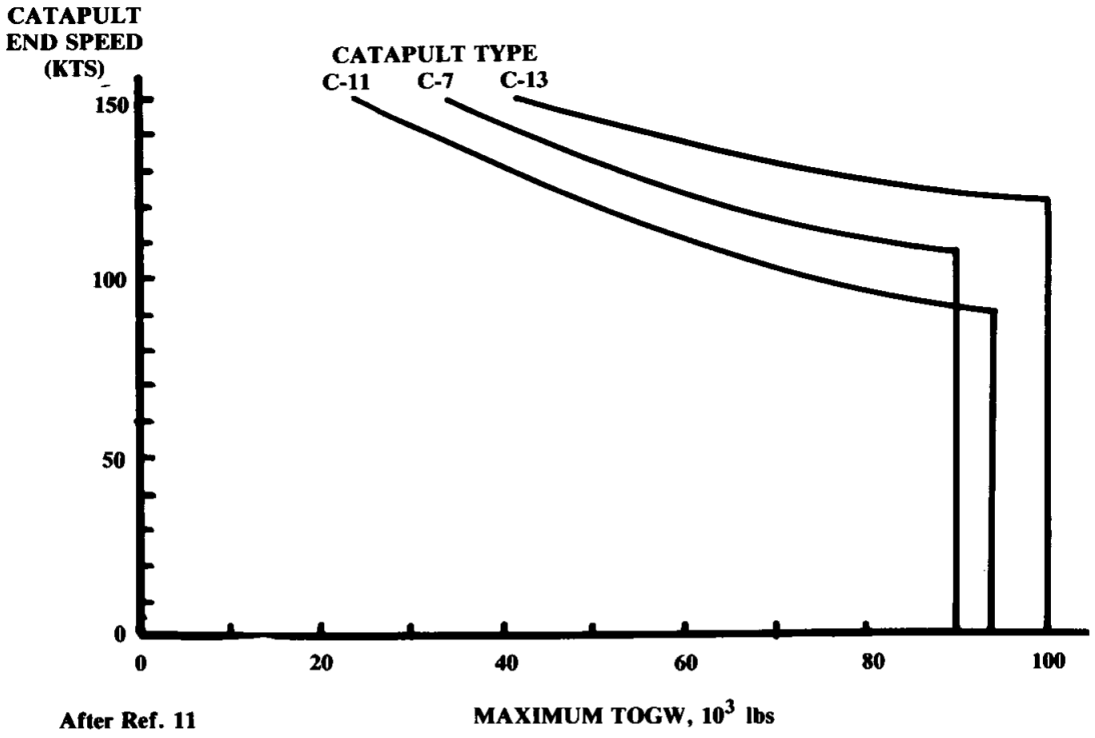


Fig. 5.5 Catapult end speeds (Raymer)

Wing Loading: Catapult Take-off

Catapult Take-off: Wind Over the Deck

- Airspeed at the end of the stroke = catapult end speed + wind-over-deck (20–40 kt), and must exceed the stall speed by **10%**.

$$\left(\frac{W}{S}\right)_{TO} = \frac{1}{2}\rho\,(V_{\text{end}} + V_{\text{wod}})^2\,\frac{(C_{L_{\max}})_{TO}}{1.21}$$

Specifications often require launch with **zero (or negative) wind-over-deck**, e.g. at anchor: the catapult alone must give the speed.